Mantra-GSC (Kors et al., 2015) is a multilingual dataset annotated in biomedical NER for five languages, however we focused only on the French subset. It covers three sources (EMEA, Medline and Patents) and use two distinct annotation schemes. These sources encompass diverse types of documents, including biomedical abstracts/titles, drug labels, and patents. To maintain evaluation uniformity, we randomly divided the dataset into three subsets: 70% for training, 10% for validation, and 20% for testing.

CLISTER (Hiebel et al., 2022) is a collection of French clinical case sentence pairs used for Semantic Textual Similarity (STS) evaluation. It consists of 1,000 sentence pairs, manually annotated by multiple annotators who assigned similarity scores ranging from 0 to 5 for each pair. These individual scores were then averaged to derive a floating-point number that represents the overall similarity of the two sentences.

CAS (Grabar et al., 2018) dataset comprises 3,790 clinical cases that underwent POS tagging with 31 different classes, using automatic tagging through the Tagex tool[4], achieving a 98% precision rate in comparison to manual annotations. This dataset involves tasks like classifying clinical cases for negation and uncertainty, as well as named-entity recognition for identifying markers of negations and speculation within medical histories and patient care. To create subsets, a random split was applied, allocating 70% for training, 10% for validation, and 20% for testing since predefined subsets were not provided.

ESSAI (Dalloux et al., 2021) consists of 7,247 clinical trial protocols that have been annotated with 41 POS tags using the TreeTagger tool (Schmid, 1994). It does also contain a classification and two named-entity recognition tasks similar to those from CAS dataset. As the dataset was not initially separated into three distinct subsets, we opted to apply the same processing methodology as we did for CAS dataset.

PxCorpus (Kocabiyikoglu et al., 2022) is a dataset designed for spoken language understanding in the medical domain, specifically focusing on transcripts related to drug prescriptions. It comprises 4 hours of transcribed dialogues, amounting to 1,981 recordings. These dialogues have been meticulously transcribed and semantically annotated. The primary task involves categorizing the textual utterances into one of four intent classes (prescribe, replace, negate, none). The second task pertains to NER, where each word in a sequence is classified into one of 38 classes, including categories such as drug, dose, or mode.

[4] https://allgo.inria.fr/app/tagex

4.2. Tokenizers Data Sources

To ensure a fair and comprehensive comparison of training data sources used by the statistical tokenizers, we carefully curated a 1GB subset of raw, lowercase text data from a variety of sources, including NACHOS (Labrak et al., 2023), PubMed Central, CC100 (Wenzek et al., 2020), and the French Wikipedia. We then constructed tokenizers using both tokenization algorithms, resulting in a total of 16 tokenizers: 8 with the integration of morphemes and 8 without. These specific data sources were chosen for their diversity: NACHOS focuses on French biomedical content, PubMed Central on English biomedical content, Wikipedia on general French language, and CC100 on general multilingual content. Each tokenizer was configured with a vocabulary size of 32k tokens, consistent with the original hyperparameters used in other French biomedical models such as CamemBERT-BIO (Touchent et al., 2023) and DrBERT (Labrak et al., 2023).

4.3. Language Model Pre-Training

To assess the impact of introducing morphemes into tokenizers on the pre-training process of biomedical language models, we conducted pre-training from scratch using the 16 tokenizer combinations (see Section 4.2). Our choice of architecture was RoBERTa (Liu et al., 2020), which is based on the masked language modeling objective and configured with standard token masking percentages as introduced by the authors.

For the PLMs training data, we utilized the NACHOS corpus created by (Labrak et al., 2023). This corpus, already pre-processed and converted to lowercase, is consistent with the data sources used for training the tokenizers. It comprises 1.1 billion words, equivalent to 7.4GB of raw text data, sourced from a wide range of online resources focusing on the French biomedical and clinical domains.

The pre-training process was conducted uniformly across all models, employing the same hyperparameters and executed over a 20-hour period. We harnessed the computational power of 32 V100 32GB GPUs available on the Jean-Zay supercomputer for this purpose. By maintaining consistent procedures and employing a fixed seed to mitigate randomness during training, we ensured the reliability and reproducibility of our experiments.

			BPE								SentencePiece							
			NACHOS		***PubMed***		***CC100***		***Wiki***		***NACHOS***		***PubMed***		***CC100***		***Wiki***	
Dataset	**Task**	**Metric**	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*
CAS	***CLS***	F1	94.2*	94.9	94.7	94.2*	95.2	**95.3**	94.8	94.8	94.8	94.7*	93.4**	93.6**	94.4	94.1	95.3*	95.1**
	NER Neg	SeqEval	**87.0**	83.3*	82.4**	81.3**	84.9	84.2	84.7	84.5*	86.1	86.4	83.6*	83.9	85.4	84.2**	85.6	83.2
	NER Spec	SeqEval	30.3*	30.6	35.0	28.2*	34.6	32.0	34.4	34.0	**36.1**	29.8	28.4*	22.2**	31.9	28.7*	32.1	27.0*
	POS	SeqEval	97.0**	96.9**	97.1	96.9**	97.1*	97.0**	**97.2**	96.9**	97.1*	97.0*	96.9**	96.9**	97.1	97.1	97.1	97.1
PxCorpus	***CLS***	F1	94.8	94.2	93.6	93.9	94.2	94.6	93.4	93.7	**94.9**	94.1	94.8	94.1	94.8	93.7	93.7	94.5
	NER	SeqEval	95.9	95.9	95.9	95.9	96.1	96.0	96.2	95.9	96.1	96.1	96.0	96.1	95.9	96.1	**96.2**	96.1
DEFT2020	***STS***	MSE	0.71	0.71	0.64*	**0.75**	0.70	0.67	0.71	0.69	0.72	0.71	0.63**	0.63	0.70	0.67*	0.70	0.67*
	CLS	F1	**91.0**	85.9	57.6**	73.7	79.5	76.3	77.1	66.0	83.0	85.3	80.9	66.7**	61.1*	66.3*	75.0*	77.4*
MORFITT	***CLS***	F1	68.6**	68.0**	66.5**	65.9**	68.4**	67.0**	68.7	67.3**	**69.6**	68.8*	66.8**	66.2**	68.2	67.5**	69.1**	67.7**
E3C	***NER Clinical***	SeqEval	**54.2**	53.1	52.4	48.6**	52.7	51.3**	51.1*	52.0*	54.2	52.4	52.1	51.1**	53.8	52.5*	53.2	51.7
	NER Temporal	SeqEval	82.0	81.2	80.9**	80.0**	81.8	81.2	**82.3**	80.6**	82.1	81.6	80.3**	79.8**	80.6**	81.1**	81.6*	81.73*
CLISTER	***STS***	MSE	0.63*	0.63	0.63	0.60**	0.65	0.63	0.62**	**0.66**	0.61*	0.64	0.61**	0.62**	0.62	0.60*	0.64**	0.63**
DEFT2021	***NER***	SeqEval	60.3	59.0**	58.1**	56.2**	59.4**	59.2**	60.1**	59.1**	**61.3**	60.1*	57.0**	56.6**	59.2**	59.9**	59.3**	58.9**
	CLS	F1	32.9	34.5*	33.4	32.3	34.5*	33.9	34.2	32.9	34.3	33.1	34.3	33.1	31.0	31.9*	34.2	**34.9**
ESSAI	***NER Spec***	SeqEval	60.5	60.9	56.4*	59.2	57.9	61.5	63.6	57.4	63.9	62.8	57.6	55.7*	**64.6**	62.0	61.4	63.1
	POS	SeqEval	98.4*	98.3	98.3	98.2**	98.4	98.4	98.3	98.3	98.4	98.4	98.3	98.2*	**98.4**	98.3	98.3*	98.3
	NER Neg	SeqEval	83.0	83.4	79.3	76.4	82.2	83.2	81.8	84.2*	81.3	84.0*	80.2	81.1	83.2	**84.2**	82.1	79.6*
	CLS	F1	97.3	97.1*	97.4	96.6**	97.4	96.7**	97.4	97.0**	97.3	97.3	**97.5**	97.2*	97.0	97.0	97.5*	97.0*
QUAERO	***NER Medline***	SeqEval	57.7	56.2**	55.4**	53.6**	57.9	55.0**	57.3	56.4**	**58.2**	55.5**	54.8**	52.9**	57.5*	55.8**	56.9	54.9**
	NER EMEA	SeqEval	65.6	65.1	63.9	63.1**	62.1**	62.7*	63.1**	62.6*	65.5	**65.9**	62.6**	63.8*	62.8**	63.1*	62.7*	62.0**
MantraGSC	***NER EMEA***	SeqEval	60.9	63.9	58.2*	60.6*	**69.3**	63.0	61.9*	62.3**	66.9	62.5*	56.8**	60.3	60.8*	59.5	64.0*	63.9**
	NER Medline	SeqEval	41.4*	42.9	39.3	36.2**	44.3	41.2	43.8	40.8*	41.9	39.5*	36.4**	37.8	46.4*	39.9	**47.1**	36.1*
	NER Patents	SeqEval	52.1*	53.3*	**57.0**	50.2*	57.0	53.9	53.6	52.3*	52.0	49.6*	50.7**	49.4	52.8*	48.0	50.6*	47.8*
Average performances per tasks																		
CLS		F1	**79.80**	79.10	73.87	76.10	78.20	77.30	77.60	75.28	78.98	78.88	77.95	75.15	74.42	75.08	77.47	77.77
NER		SeqEval	63.92	63.75	62.63	60.73	64.63	63.42	64.15	63.24	**65.05**	63.55	61.27	60.82	64.22	62.69	64.06	62.00
POS		SeqEval	97.70	97.60	97.70	97.55	**97.75**	97.70	97.75	97.60	97.75	97.70	97.60	97.55	97.75	97.70	97.70	97.70
STS		MSE	0,67	0,67	0,64	0,68	**0,68**	0,65	0,67	0,68	0,67	0,68	0,62	0,63	0,66	0,64	0,67	0,65

Table 2: Performance of the tokenization algorithms and different data sources used to train tokenizers (top). Average performance per type of tasks is also reported (bottom). *w/o* and *w/* denote models without and with morphemes. Best models are in bold, and the second-best are underlined. Statistical significance is determined using Student's t-test, where * indicates $p < 0.05$, and ** $p < 0.01$.

4.4. Evaluation

All models undergo fine-tuning following a standardized protocol with identical hyperparameters for each downstream task, enabling a focused evaluation of tokenizers. We ensure robustness and reliability by averaging the results across four independent runs and performing statistical significance assessments using Student's t-test.

For consistent comparisons, especially in sequence-to-sequence tasks like POS tagging and NER, we employ the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format. To align with established practices (Touchent et al., 2023), our models are trained to predict only the label for the initial token of each word.

5. Results and Discussions

In this section, we present the results of our tokenization strategies on various biomedical NLP tasks, with a focus on key aspects. We investigate the impact of tokenization granularity (Section 5.1), the introduction of morphological information during tokenizer construction (Section 5.2), and the influence of data sources on tokenizers, including token sparsity, morpheme coverage, and the overall performance of different tokenization algorithms (Section 5.3).

Table 2 summarizes the performance of the BPE and SentencePiece strategies, both with (*w/*) and without our morpheme-enriched approach (*w/o*), across various French biomedical downstream tasks. Average performance per task type is also provided for clarity. It's worth noting that, before delving into detailed analysis, there is no consistent tokenization strategy that consistently yields the best results in all tasks, whether it employs a purely statistical algorithm or a statistical approach coupled with morpheme enrichment.

5.1. Impact of tokenization granularity

To assess the impact of tokenization granularity, Table 3 presents the average number of sub-word units per word for each tokenization strategy and data source used in the studied tasks. While deriving overarching conclusions from these results can be challenging, we calculated Pearson correlation (ρ) between models performances on the downstream tasks from Table 2 and the corresponding average number of sub-word units per word. These correlation scores range from -1 to $+1$, where -1 indicates a complete negative linear correlation, 0 represents no correlation, and $+1$ signifies a strong positive correlation. In the context of tokenization,

		BPE								SentencePiece								
		NACHOS		*PubMed*		*CC100*		*Wikipedia*		*NACHOS*		*PubMed*		*CC100*		*Wikipedia*		
Corpus	**Task**	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	*w/o*	*w/*	ρ
CAS	**CLS**	1.32	1.38	2.20	2.13	1.49	1.49	1.51	1.50	1.32	1.45	2.18	2.15	1.49	1.56	1.51	1.57	*-0.62*
	NER Neg	1.32	1.38	2.20	2.13	1.49	1.49	1.51	1.50	1.32	1.45	2.18	2.15	1.49	1.56	1.51	1.57	*-0.70*
	NER Spec	1.32	1.38	2.20	2.13	1.49	1.49	1.51	1.50	1.32	1.45	2.18	2.15	1.49	1.56	1.51	1.57	*-0.42*
	POS	1.32	1.38	2.20	2.13	1.49	1.49	1.51	1.50	1.32	1.45	2.18	2.15	1.49	1.56	1.51	1.57	*-0.36*
PxCorpus	**CLS**	1.54	1.62	2.26	2.27	1.76	1.72	1.73	1.72	1.54	1.67	2.24	2.30	1.72	1.77	1.77	1.82	*-0.22*
	NER	1.54	1.62	2.26	2.27	1.76	1.72	1.73	1.72	1.54	1.67	2.24	2.30	1.72	1.77	1.77	1.82	*-0.22*
DEFT2020	**STS**	1.41	1.45	2.27	2.24	1.42	1.45	1.43	1.45	1.41	1.49	2.24	2.23	1.41	1.48	1.42	1.49	*-0.47*
	CLS	1.21	1.26	2.13	2.09	1.31	1.34	1.33	1.36	1.20	1.32	2.05	2.04	1.25	1.34	1.29	1.37	*-0.41*
MorFITT	**CLS**	1.38	1.44	2.45	2.40	1.48	1.50	1.49	1.51	1.37	1.50	2.35	2.33	1.46	1.55	1.48	1.57	*-0.82*
E3C	**NER Clinical**	1.30	1.35	2.23	2.17	1.48	1.48	1.50	1.49	1.29	1.43	2.22	2.18	1.48	1.55	1.49	1.56	*-0.59*
	NER Temporal	1.29	1.35	2.22	2.16	1.48	1.48	1.48	1.49	1.29	1.43	2.22	2.18	1.47	1.54	1.48	1.55	*-0.75*
CLISTER	**STS**	1.52	1.59	2.65	2.57	1.73	1.72	1.74	1.72	1.51	1.65	2.56	2.49	1.71	1.77	1.71	1.77	−0.33
DEFT2021	**NER**	1.31	1.37	2.26	2.19	1.48	1.49	1.50	1.50	1.31	1.44	2.19	2.15	1.48	1.55	1.49	1.56	*-0.88*
	CLS	1.50	1.57	2.63	2.56	1.69	1.70	1.71	1.71	1.46	1.61	2.50	2.46	1.64	1.72	1.66	1.74	*-0.11*
ESSAI	**NER Spec**	1.29	1.34	2.20	2.14	1.42	1.43	1.45	1.45	1.29	1.41	2.21	2.16	1.41	1.49	1.46	1.52	*-0.68*
	POS	1.28	1.33	2.19	2.13	1.41	1.42	1.44	1.44	1.28	1.41	2.19	2.15	1.40	1.48	1.44	1.51	*-0.61*
	NER Neg	1.28	1.33	2.19	2.13	1.41	1.42	1.44	1.44	1.28	1.41	2.19	2.15	1.40	1.48	1.44	1.51	*-0.69*
	CLS	1.28	1.34	2.20	2.14	1.42	1.43	1.45	1.46	1.28	1.41	2.20	2.16	1.41	1.49	1.45	1.52	*-0.02*
QUAERO	**NER Medline**	1.53	1.63	2.35	2.26	1.78	1.78	1.77	1.78	1.52	1.76	2.36	2.35	1.77	1.89	1.76	1.89	*-0.77*
	NER EMEA	1.30	1.34	2.14	2.12	1.44	1.46	1.49	1.51	1.30	1.39	2.06	2.04	1.45	1.51	1.50	1.56	*-0.28*
MANTRAGSC	**NER EMEA**	1.33	1.40	2.47	2.41	1.49	1.51	1.50	1.52	1.32	1.43	2.33	2.30	1.46	1.53	1.49	1.55	*-0.63*
	NER Medline	1.89	2.01	2.84	2.70	2.06	2.13	2.14	2.14	1.89	2.09	2.84	2.78	2.06	2.22	2.10	2.22	*-0.64*
	NER Patents	1.54	1.59	2.34	2.30	1.61	1.63	1.59	1.62	1.43	1.52	2.20	2.20	1.50	1.58	1.51	1.60	*0.06*
Average per model		1.39	1.45	2.30	2.25	1.54	1.55	1.56	1.56	1.38	1.51	2.26	2.24	1.52	1.60	1.55	1.62	*-0.48*
Relative Difference (%)		0.0	4.5	65.9	61.8	11.2	11.8	12.3	12.6	−0.7	8.9	62.8	61.1	9.9	15.5	11.7	16.9	

Table 3: Average number of sub-word units per word for each tokenization strategy and data source training. Their Pearson correlation (ρ) with each task performance is reported (last column). Cells colored in red correspond to lower performing models, while those in green represent higher ones. The last row represents the relative difference in terms of average subwords per word compared to the NACHOS BPE without morpheme baseline. *w/o* and *w/* denote models without and with morphemes.

a negative correlation implies that fewer subword units are associated with higher scores, while a positive correlation suggests that more subword units are linked to higher scores.

In overall, we observe in Table 3 an average ρ correlation of -0.48 between tasks and models, indicating that, in general, higher performance scores tend to be associated with fewer subword units. To our knowledge, this is the first time such a correlation has been experimentally demonstrated. However, it's important to note that this correlation varies across the targeted tasks. Tasks like CLS show correlation close to zero, suggesting that they are less affected by the granularity of tokenization. In contrast, STS and sequence-to-sequence tasks, particularly NER, appear to be more influenced by tokenization granularity, likely due to their heavy reliance on immediate context for making predictions.

While the RoBERTa model's embeddings capture semantic meaning and the encoder module captures contextual information (Rogers et al., 2020), we aimed to determine whether the observed correlations are attributed to a specific part of this architecture. To investigate this, we isolated and froze the embeddings and/or encoder of our BERT-based model, based on the NACHOS SentencePiece, during fine-tuning for various tasks. The experimental approach, as detailed in Table 4, involved several stages. Initially, we established a baseline for each task with no frozen components. Subsequently, we conducted experiments by freezing only the embedding layer, only the encoder, and both the embeddings and encoders. Our findings indicate a stronger dependence on RoBERTa's encoder for tasks such as POS tagging and STS, in contrast to other tasks, which corroborate the context dependency as an explanation to the correlation scores between segmentation granularity and models performances for these tasks but not for NER.

	CAS POS	PxCorpus NER	PxCorpus CLS	CLISTER STS
🔥 **Full Fine-tuning**	97.10	96.10	94.82	0.61
❄ **Embedding**	97.03 ↓ 0.07	96.10 ↑ 0.00	94.73 ↓ 0.09	0.62 ↑ 1.63
❄ **Encoder**	65.97 ↓ 32.05	83.95 ↓ 12.64	84.78 ↓ 10.58	0.45 ↓ 26.22
❄ **Embedding + Encoder**	60.04 ↓ 38.16	79.62 ↓ 17.14	84.78 ↓ 10.58	0.44 ↓ 27.86

Table 4: Performance and relative loss (in %) of the PLMs based on SentencePiece NACHOS without morpheme with parts of the models being frozen.

As shown in Table 3, higher performance scores are associated with fewer subword units. To gain a linguistic perspective on how tokenization strategies behave, we analyzed the segmentation of 150 biomedical terms equally distributed across cardiology, dermatology, obstetric-gynecology, and